

Fusion draft: SST-11/41 Water and time cluster

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About this fusion draft

This is an automatically assembled fusion draft: multiple SST manuscripts concatenated as chapters. It is intended for navigation and editing workflow, not as a final merged publication without manual cleanup.

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Chapter 1

SST-41 (survivor) — Fluid fine-structure

Vorticity Generation, Angle Reversal, and a Fluid Fine-Structure Analogy

Abstract

We investigate how a rapidly rotating, incompressible fluid contained in a cylinder responds when a neutrally buoyant spherical control volume on the axis is forced to move vertically. In the limit of small Rossby and Ekman numbers, linear rotating-fluid theory reduces to a compact vorticity-production law,

$$\partial_t \omega_z = 2\Omega \partial_z w,$$

This relation predicts vertical vorticity of opposite sign above and below the driver, leading to instantaneous tracer rotation in opposite directions. For symmetric up–down strokes, the accumulated angle reverses on the return path, so that the net rotation over a full cycle cancels at leading order.

As a speculative extension, we introduce a dimensionless “fluid fine-structure constant” that connects swirl speed to a tracer’s clock-rate rule. Unlike the angle response, this hypothesis predicts a small but non-reversing cycle-averaged proper-time deficit. We present closed-form estimates, dimensional checks, and limiting cases, while stressing that the analogy is offered as a testable conjecture, separate from the rigorous macroscopic results.

1.1 Physical setting and asymptotic regime

A cylinder of radius R and height H contains an incompressible fluid of density ρ . The container rotates at constant angular speed Ω about $\hat{\mathbf{z}}$. In the rotating frame the base state is at rest; the absolute vorticity is uniform, $\boldsymbol{\omega}_a^{(0)} = 2\boldsymbol{\Omega}$ [?, ?].

A neutrally buoyant spherical control volume of radius a is centered on the axis $r = 0$ and is forced vertically with displacement $Z(t)$. The forcing is axisymmetric and smooth; denote the induced vertical velocity by $w(r, z, t)$. We work in the regime

$$\text{Ro} = \frac{U'}{\Omega L} \ll 1, \quad \text{E} = \frac{\nu}{\Omega L^2} \ll 1,$$

with perturbation speed U' and length $L = \mathcal{O}(a)$, so that linear, inviscid rotating-flow theory applies away from thin boundary layers [?, ?, ?].

1.2 Governing equations and vorticity production

In the rotating frame the inviscid equations are

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + 2\boldsymbol{\Omega} \times \mathbf{u} = -\nabla \Pi, \quad (1.1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (1.2)$$

Taking curl of (A.18) gives the absolute-vorticity equation [?, ?]

$$\partial_t \boldsymbol{\omega} = \nabla \times (\mathbf{u} \times \boldsymbol{\omega}_a), \quad \boldsymbol{\omega}_a = \boldsymbol{\omega} + 2\boldsymbol{\Omega}. \quad (1.3)$$

Linearizing about $\boldsymbol{\omega}_a^{(0)} = 2\boldsymbol{\Omega}$ (neglect quadratic perturbation terms) yields

$$\partial_t \boldsymbol{\omega} \approx 2(\boldsymbol{\Omega} \cdot \nabla) \mathbf{u}. \quad (1.4)$$

Axisymmetry implies that the vertical component obeys

$$\boxed{\partial_t \omega_z = 2\Omega \partial_z w.} \quad (1.5)$$

Introduce a vertical displacement field $\xi(r, z, t)$ with $w = \partial_t \xi$. Integrating (A.22) from an unperturbed initial state,

$$\boxed{\omega_z(r, z, t) = 2\Omega \partial_z \xi(r, z, t).} \quad (1.6)$$

Equation (A.23) is the rotating analogue of vortex-line stretching: regions of column stretching ($\partial_z \xi > 0$) generate cyclonic vorticity, while compression ($\partial_z \xi < 0$) generates anticyclonic vorticity [?, ?].

1.3 From vertical vorticity to azimuthal velocity

Under axisymmetry the kinematic relation between vertical vorticity and azimuthal velocity is

$$\omega_z(r, z, t) = \frac{1}{r} \partial_r (r u_\theta(r, z, t)). \quad (1.7)$$

Regularity at $r = 0$ ($u_\theta \sim r$) gives

$$\boxed{u_\theta(r, z, t) = \frac{1}{r} \int_0^r \omega_z(r', z, t) r' dr' = \frac{2\Omega}{r} \int_0^r \partial_z \xi(r', z, t) r' dr'.} \quad (1.8)$$

1.3.1 A concrete smooth kernel

Let

$$\xi(r, z, t) = Z(t) \psi(r, z), \quad \psi(r, z) = \exp\left(-\frac{r^2 + z^2}{a^2}\right). \quad (1.9)$$

Then $\partial_z \xi = -\frac{2Z(t)z}{a^2} \psi$, and (A.24) yields

$$\boxed{u_\theta(r, z, t) = -\frac{2\Omega Z(t)z}{a^2} e^{-z^2/a^2} \frac{1 - e^{-r^2/a^2}}{r}.} \quad (1.10)$$

Sign structure: for $Z(t) > 0$ (up-stroke), $u_\theta \propto -z$, so the azimuthal response is cyclonic below ($z < 0$) and anticyclonic above ($z > 0$).

Near the axis, $1 - e^{-r^2/a^2} \sim r^2/a^2$, hence $u_\theta \sim -\frac{2\Omega Z(t)z}{a^4} e^{-z^2/a^2} r$ (regular).

1.4 Angle reversal and reversibility

Define the relative angular rate $\dot{\theta}_{\text{rel}} = u_\theta/r$. Over a stroke from t_1 to t_2 ,

$$\Delta\theta_{\text{rel}}(r, z) = \int_{t_1}^{t_2} \frac{u_\theta}{r} dt. \quad (1.11)$$

Because $\dot{\theta}_{\text{rel}}$ is *linear* in $Z(t)$, a symmetric up–down cycle with zero mean displacement satisfies

$$\boxed{\Delta\theta_{\text{rel}}(r, z; \text{one period}) = 0 \quad (\text{to leading order in Ro, E}).} \quad (1.12)$$

This is the expected quasi-static reversibility of linear, rapidly rotating flow (the Taylor–Proudman framework) [?, ?, ?]. Deviations arise at higher order from viscosity (Ekman pumping), finite-amplitude advection (steady streaming), or near-resonant inertial waves [?, ?].

1.5 A fluid-inspired kinematic time hypothesis

We now explore a speculative analogy, motivated by the parity property of swirl cancellation. The macroscopic fluid equations remain unchanged; the following is presented only as a *dimensionless coupling hypothesis*.

1.5.1 Definition of a fluid fine-structure constant

Introduce a microscopic length scale equal to half the classical electron radius,

$$L \equiv \frac{1}{2}r_e, \quad r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2} \approx 2.82 \times 10^{-15} \text{ m}.$$

With vorticity related to swirl velocity by $\omega = 2u_\theta/C_e$, we define the *fluid fine-structure constant*

$$\boxed{\alpha_f \equiv \frac{\omega L}{c} = \frac{r_e}{c C_e} u_\theta.} \quad (1.13)$$

This quantity is dimensionless, and—like the electromagnetic fine-structure constant—it measures the relative strength of a coupling (here, swirl to clock-rate).

1.5.2 Time-rate rule

We propose that the local tracer clock rate is

$$\boxed{\frac{d\tau}{dt} = \sqrt{1 - \alpha_f^2}} \quad (1.14)$$

For $\alpha_f \ll 1$, expansion yields

$$\frac{d\tau}{dt} \approx 1 - \frac{1}{2}\alpha_f^2 + \mathcal{O}(\alpha_f^4).$$

Thus the leading correction is quadratic in α_f , so angle reverses on a symmetric stroke but the time deficit does not—a clear falsifiable signature.

1.5.3 Cycle-averaged deficit

For sinusoidal forcing $Z(t) = Z_0 \sin \sigma t$,

$$\Delta\tau = -\frac{1}{2}\alpha_f^2 T + \mathcal{O}(\alpha_f^4), \quad T = \frac{2\pi}{\sigma}.$$

With representative laboratory parameters ($\Omega \sim 2$ rad/s, $a \sim 5$ cm, $Z_0 \sim 2$ mm), we estimate $\alpha_f \sim 10^{-8}$ – 10^{-9} , giving fractional time-rate shifts of order 10^{-16} per cycle.

1.6 Discussion and experimental considerations

The macroscopic prediction of opposite-sign swirl and cycle reversibility can be demonstrated directly with simple tracer experiments. For instance, dye or ink layers placed at $\pm z_0$ should rotate in opposite directions during a stroke, with the motions canceling once the driver returns. Such a setup offers a clear and even classroom-level illustration of angle cancellation in rotating flows.

The speculative time-rate analogy points to a different outcome: a cycle-averaged deficit of order α_f^2 that does not reverse. Detecting such a subtle effect would require synchronized tracer clocks capable of fractional timing resolution on the order of 10^{-16} per cycle—far beyond current experimental capabilities, but in principle measurable. Even a null result would be informative, since it would place bounds on α_f and therefore limit the analogy.

Conclusion

We have analyzed the azimuthal response of a vertically forced, axisymmetric driver in a rapidly rotating fluid. The theory predicts opposite-sign vertical vorticity above and below the driver, producing tracer motions that cancel over a symmetric cycle. This reversibility is consistent with classical rotating-flow dynamics in the low-Rossby, low-Ekman regime.

As a complementary exploration, we introduced a dimensionless *fluid fine-structure constant* α_f together with a kinematic time-rate rule. While the resulting prediction—a minuscule but non-reversing time deficit quadratic in α_f —remains far below present experimental resolution, it is nonetheless falsifiable in principle. Framed in this way, the analogy connects the present work to the analogue-gravity tradition while keeping a clear distinction between established macroscopic dynamics and speculative microscopic interpretation.

.1 Details for the Gaussian kernel

With $\psi = \exp(-(r^2 + z^2)/a^2)$,

$$\partial_z \xi = Z(t) \partial_z \psi = -\frac{2Z(t)z}{a^2} e^{-(r^2+z^2)/a^2}.$$

Inserting into (A.24):

$$u_\theta(r, z, t) = \frac{2\Omega}{r} \int_0^r \left(-\frac{2Z(t)z}{a^2} e^{-(r'^2+z^2)/a^2} \right) r' dr'.$$

The radial integral evaluates to

$$\int_0^r e^{-r'^2/a^2} r' dr' = \frac{a^2}{2} \left(1 - e^{-r^2/a^2} \right),$$

giving the stated result (A.25). The near-axis expansion follows from $1 - e^{-r^2/a^2} \sim r^2/a^2$.

.2 When reversibility can fail (order estimates)

Let $\epsilon \sim \text{Ro}$ be a small parameter. Viscous corrections scale as $\mathcal{O}(\text{E}^{1/2})$ in bulk via Ekman pumping. Quadratic advection $(\mathbf{u} \cdot \nabla) \mathbf{u}$ introduces steady streaming at $\mathcal{O}(\epsilon^2)$, yielding a nonzero mean angle per cycle. Near the inertial band $|\sigma - 2\Omega| \ll 2\Omega$, wave radiation produces phase lags $\mathcal{O}(\epsilon)$ that also break exact reversal [?]. Operating with $\sigma \ll 2\Omega$, $\epsilon \ll 1$, and small E ensures the leading-order predictions.

Appendix A

SST-11 — Water and Time

**Impulsive Axisymmetric Forcing in a Rotating Cylinder,
Reversible Swirl Response, and Skymionic Photon Emission:
Fluid Benchmarks and Fluid-Inspired Kinematic Hypotheses**

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Abstract

We present a one-to-one Swirl–String Theory (SST) translation of a rotating-tank impulse experiment and skyrmionic-photon framework. All sections/equations from the original are preserved, with SST terms added via Rosetta mapping and mainstream physics glosses inline.

Macrodynamics remain standard rotating Euler; SST introduces (i) bookkeeping of swirl energy/mass densities and (ii) the canonical *Swirl Clock* $d\tau/dt = \sqrt{1 - v^2/c^2}$ alongside the legacy $\sqrt{1 - u^2/C_e^2}$ hypothesis (kept as an explicit, testable variant). Dimensional checks and numerics use $\|\mathbf{v}_\odot\| = 1.09384563 \times 10^6 \text{ m s}^{-1}$, $r_c = 1.40897017 \times 10^{-15} \text{ m}$, $\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}$.

Rosetta card (SST $\leftrightarrow$ mainstream)

$$\begin{aligned}
 \rho_E &= \frac{1}{2} \rho_f \|\mathbf{v}\|^2 && (\text{mainstream: kinetic-energy density}), \\
 \rho_m &= \rho_E / c^2 && (\text{mainstream: mass from energy}), \\
 \Omega_0 &\equiv \|\mathbf{v}_\zeta\| / r_c && (\text{SST characteristic frequency}), \\
 S_t &= \sqrt{1 - v^2 / c^2} && (\text{Swirl Clock; mainstream: Lorentz factor}).
 \end{aligned}$$

Distinct roles: Ω (lab rotation) controls inertial waves; Ω_0 is an SST material scale derived from $\mathbf{v}_\zeta$ and r_c .

A.1 Set-up and observation

Consider a vertical cylinder of radius R and depth H , rotating at rate Ω about z . A bottom DC motor drives a three-fin impeller that briefly produces a hollow-core vortex of radius $a_v \approx 7.5$ mm and axial jetting. After a delay $\sim H/c_{g,z}$, a two-lobed "push-pull" on-axis surface signal appears at $z = H$.

SST map. Energy in the packet that reaches the surface is tracked as $\rho_E = \frac{1}{2} \rho_f \|\mathbf{u}'\|^2$ and mass-equivalent density $\rho_m = \rho_E / c^2$ (bookkeeping only).

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A.2 Linear rotating-wave framework

In the bulk, small perturbations admit *inertial waves* with dispersion[?, ?, ?]

$$\omega = 2\Omega \frac{k_z}{k}, \quad k = \sqrt{k_r^2 + k_z^2}. \quad (\text{A.1})$$

For axisymmetry ($m = 0$) in a cylinder, $k_r \approx \lambda_{0n}/R$ (Bessel eigenvalues)[?]. Group velocities follow from $\partial\omega/\partial k_i$:

$$c_{g,r} = -\frac{2\Omega k_z k_r}{k^3}, \quad c_{g,z} = \frac{2\Omega k_r^2}{k^3}. \quad (\text{A.2})$$

Beams satisfy $\tan \alpha = |c_{g,r}|/c_{g,z} = k_z/k_r$.

Arrival time. With $c_{g,z}$ as above,

$$t_{\text{arr}} \approx \frac{H}{c_{g,z}} = \frac{H k^3}{2\Omega k_r^2}. \quad (\text{A.3})$$

Numerics (bench-top). $R = 0.075 \text{ m}$, $H = 0.30 \text{ m}$, $\Omega \approx 2 \text{ rad/s}$, $k_r \approx 3.83/R \approx 51 \text{ m}^{-1}$, $k_z \approx \pi/H \approx 10.5 \text{ m}^{-1}$. Then $k \approx 52.1 \text{ m}^{-1}$, $c_{g,z} \approx 7.4 \times 10^{-2} \text{ m/s}$, $t_{\text{arr}} \approx 4.1 \text{ s}$. Units: $[H/c_{g,z}] = \text{s}$.

Analogy (10-year-old). A tilted line of dominos falls in a slanted path; it reaches the top after a short delay.

A.3 Impulse sign and the observed "push–pull"

Start impulse: axial upwelling and centrifugal low p' in the hollow core yield a *surface depression* on arrival. Stop impulse reverses the sign and gives a *surface rise*. Hydrostatic surface coupling[?, ?]:

$$p'(z=0) + \rho g \eta = 0 \Rightarrow \eta = -\frac{p'(0)}{\rho g}. \quad (\text{A.4})$$

A.4 Relation to vortex rings and jet starting vortices

A short burst sheds a starting ring with translational speed in quiescent fluid[?]

$$U_{\text{ring}} \approx \frac{\Gamma}{4\pi R_v} \left[\ln\left(\frac{8R_v}{a_v}\right) - \frac{1}{4} \right]. \quad (\text{A.5})$$

In rotation, the on-axis *delayed* packet at the surface is dominated by the inertial-wave field, not by ring ballistic motion.

A.5 What this *is* and *is not* an analogy to

Photon analogy (limited)

Localized impulse $\Rightarrow$ localized packet with clear parity. Inertial waves are anisotropic and dispersive ((A.1)–(A.2)); free photons in vacuum are isotropic and nondispersive. Analogy is qualitative.

Gravitational-wave analogy (limited)

GR waves are transverse, quadrupolar, nondispersive (in vacuum). Axisymmetric inertial-wave response is neither quadrupolar nor nondispersive; c_g depends on (k_r, k_z) and Ω .

A.6 A proposed microscopic interpretation inspired by fluid analogs (VAM)

Legacy rule (kept as testable hypothesis). Let $u(r, t)$ be local swirl speed. Postulate

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{u^2}{C_e^2}}, \quad (\text{A.6})$$

with C_e a characteristic speed. (Rosetta: $C_e \mapsto \|\mathbf{v}_O\|$.)

Canonical SST (Swirl Clock). SST uses

$$\frac{d\tau}{dt} = S_t = \sqrt{1 - \frac{v^2}{c^2}} \quad (v = |u_\theta| \text{ locally}), \quad (\text{A.7})$$

recovering Lorentz kinematics.

Parity prediction. Two opposite-sign swirl impulses flip surface angle, but any time-rate effect even in speed ((A.6) or (A.7) expansion) does not reverse.

A.7 Falsifiable checks

- **Travel-time scaling:** $t_{\text{arr}} \propto \Omega^{-1}$; infer (k_r, k_z) from beam angle or cylinder modes[?].
- **Bipolarity:** Start/stop impulses give opposite η via (A.4).
- **Anisotropy:** Off-axis probes detect beams at angle $\tan \alpha = k_z/k_r$; axis receives vertical arrival.
- **Time-rate parity:** Two "clock tracers" at the surface (on-axis vs off-axis reference); bound even-in-speed time drift.

Conclusion (Part I)

An impulsive rotor in a rotating cylinder launches an axisymmetric inertial-wave packet whose $c_{g,z}$ yields a delayed on-axis surface response of the observed magnitude. The two-lobed signal follows the impulse signs. A speed-based time-rate rule, whether legacy (A.6) or canonical (A.7), produces even-in-speed effects that do not reverse.

A.8 Why the Surface "Push–Pull" Requires Background Rotation

Observation. The delayed bipolar surface signal occurs only for nonzero Ω ; for $\Omega = 0$ there is no reproducible delayed response.

A.8.1 Mechanism: Coriolis restoring force and inertial-wave beams

Linear rotating Euler equations[?]:

$$\partial_t \mathbf{u} + 2\mathbf{\Omega} \times \mathbf{u} = -\nabla \pi, \quad \nabla \cdot \mathbf{u} = 0. \quad (\text{A.8})$$

They admit inertial waves with dispersion and group velocity ((A.1)–(A.2)). Arrival time (A.3). Hydrostatic coupling (A.4).

A.8.2 Scaling explaining the "off" state at $\Omega = 0$

As $\Omega \rightarrow 0$: $\omega \rightarrow 0$, $c_{g,z} \propto \Omega \rightarrow 0$, $t_{\text{arr}} \rightarrow \infty$. Geostrophic scaling yields $\eta \propto \Omega$; viscous damping also grows (Ekman). Hence the delayed bipolar signature vanishes.

A.8.3 Testable predictions vs Ω

$$t_{\text{arr}}(\Omega) \propto \Omega^{-1}, \quad (\text{A.9})$$

$$\eta_{\text{max}}(\Omega) \propto \Omega. \quad (\text{A.10})$$

Dimensional checks: $[H/c_{g,z}] = \text{s}$; $[p'/\rho g] = \text{m}$.

A.9 Hypothesized Compressional Signaling Branch (Fluid-Inspired, Microscopic)

Summary. Add a longitudinal compressional channel with speed $c_P \gg c$ that couples (ultra-weakly) to isotropic stress. Macrodynamics unchanged.

A.9.1 Minimal linear model

Relax strict incompressibility and let $p' = B_* \rho' / \rho_*$; linearization gives[?, ?]

$$\partial_t^2 \rho' - c_P^2 \nabla^2 \rho' = 0, \quad c_P = \sqrt{\frac{B_*}{\rho_*}}. \quad (\text{A.11})$$

Dimension: $[c_P] = \text{m s}^{-1}$.

A.9.2 Field-theoretic sketch (source coupling)

Introduce scalar Φ with action

$$S_\Phi = \int dt d^3x \frac{1}{2} [\kappa_\Phi^{-1} (\partial_t \Phi)^2 - \Lambda_\Phi^2 (\nabla \Phi)^2] + \int dt d^3x \varepsilon \Phi \mathcal{T}, \quad (\text{A.12})$$

where $\mathcal{T}$ is isotropic stress trace. Euler–Lagrange equation

$$\partial_t^2 \Phi - C_*^2 \nabla^2 \Phi = \varepsilon \kappa_\Phi \mathcal{T}, \quad C_* \equiv \Lambda_\Phi \sqrt{\kappa_\Phi}. \quad (\text{A.13})$$

Identifying $\rho' \propto \partial_t \Phi$ maps $C_* \rightarrow c_P$ and (A.11).

A.9.3 Numerical checks with supplied parameters

Using $\rho_* = 7.0 \times 10^{-7} \text{ kg m}^{-3}$, $B_* \approx 3.49924562 \times 10^{35} \text{ J m}^{-3}$:

$$c_P = \sqrt{B_*/\rho_*} = 7.0703057319 \times 10^{20} \text{ m s}^{-1}. \quad (\text{A.14})$$

Ratios: $c_P/c \approx 2.36 \times 10^{12}$; for $L = 10 \text{ m}$, $t = L/c_P \approx 1.4 \times 10^{-20} \text{ s}$. Compatibility: GW170817 constrains $v_{\text{GW}} \approx c$ [?, ?]; a distinct, trace-coupled channel can evade provided coupling is tiny.

A.9.4 Falsifiable protocol

Two baselines $L_1 < L_2$; earliest correlated arrivals t_1, t_2 (after excluding EM/acoustic paths) obey

$$c_P \gtrsim \frac{L_2 - L_1}{t_2 - t_1}. \quad (\text{A.15})$$

Null at resolution δt yields $c_P \gtrsim (L_2 - L_1)/\delta t$.

A.10 Replacing the Working Fluid by a VAM-Like Superfluid Medium

Aim. Consider a hypothetical inviscid superfluid with microscopic swirl. Macroscopically, keep rotating Euler; microscopically, two branches: transverse (inertial-like) and longitudinal (compressional).

A.10.1 Transverse inertial-like branch with effective background rate

Define $\Omega_* \equiv C_e/\ell_*$. Then

$$\omega = 2\Omega_* \frac{k_z}{k}, \quad c_{g,z} = \frac{2\Omega_* k_r^2}{k^3} \sim 2C_e \frac{R}{\ell_*}. \quad (\text{A.16})$$

Arrival:

$$t_{\text{arr}} \sim \frac{H \ell_*}{2C_e R}. \quad (\text{A.17})$$

Numbers. $R = 7.5$ cm, $H = 30$ cm, $C_e = 1.09384563 \times 10^6$ m s⁻¹: (i) $\ell_* = R$: $t \sim H/(2C_e) \approx 1.37 \times 10^{-7}$ s; (ii) $\ell_* = r_c$: $t \sim 2.6 \times 10^{-21}$ s. Units: $[H\ell_*/(C_e R)] = \text{s}$.

A.10.2 Longitudinal compressional branch

As in (A.11), $c_P = \sqrt{B_*/\rho_*}$ is kinematic front speed; detection requires ultra-weak coupling to avoid conflicts with known tests[?, ?].

Part II: Reversible Azimuthal Response to Axisymmetric Vertical Forcing

Setting. Cylinder radius R , height H , fluid density ρ . Container rotates at Ω ; base state at rest in rotating frame; absolute vorticity $\omega_a^{(0)} = 2\Omega$ [?, ?].

A.10.3 Governing equations and vorticity production

Inviscid rotating-frame equations:

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + 2\boldsymbol{\Omega} \times \mathbf{u} = -\nabla \Pi, \quad (\text{A.18})$$

$$\nabla \cdot \mathbf{u} = 0. \quad (\text{A.19})$$

Curl gives absolute-vorticity equation[?, ?]

$$\partial_t \boldsymbol{\omega} = \nabla \times (\mathbf{u} \times \boldsymbol{\omega}_a), \quad \boldsymbol{\omega}_a = \boldsymbol{\omega} + 2\boldsymbol{\Omega}. \quad (\text{A.20})$$

Linearizing about $\boldsymbol{\omega}_a^{(0)}$ yields

$$\partial_t \boldsymbol{\omega} \approx 2(\boldsymbol{\Omega} \cdot \nabla) \mathbf{u}. \quad (\text{A.21})$$

Axisymmetry $\Rightarrow$ vertical component

$$\partial_t \omega_z = 2\Omega \partial_z w. \quad (\text{A.22})$$

Introduce displacement ξ with $w = \partial_t \xi$; integrate from rest:

$$\omega_z(r, z, t) = 2\Omega \partial_z \xi(r, z, t). \quad (\text{A.23})$$

A.10.4 From vertical vorticity to azimuthal velocity

Kinematic relation

$$\omega_z = \frac{1}{r} \partial_r(r u_\theta) \Rightarrow u_\theta(r, z, t) = \frac{1}{r} \int_0^r \omega_z(r', z, t) r' dr'. \quad (\text{A.24})$$

Gaussian kernel. For $\xi = Z(t) \exp(-(r^2 + z^2)/a^2)$,

$$u_\theta(r, z, t) = -\frac{2\Omega Z(t) z}{a^2} e^{-z^2/a^2} \frac{1 - e^{-r^2/a^2}}{r}. \quad (\text{A.25})$$

Sign: cyclonic below ($z < 0$), anticyclonic above ($z > 0$). Near-axis regularity: $1 - e^{-r^2/a^2} \sim r^2/a^2$.

A.10.5 Angle reversal and reversibility

Relative angular rate $\dot{\theta}_{\text{rel}} = u_\theta/r$. Over a stroke $[t_1, t_2]$,

$$\Delta\theta_{\text{rel}}(r, z) = \int_{t_1}^{t_2} \frac{u_\theta}{r} dt. \quad (\text{A.26})$$

Because $\dot{\theta}_{\text{rel}}$ is linear in $Z(t)$, a symmetric up-down cycle with zero mean displacement yields

$$\boxed{\Delta\theta_{\text{rel}}(r, z; \text{one period}) = 0 \quad (\text{to leading order}).} \quad (\text{A.27})$$

Deviations arise from viscosity (Ekman), quadratic advection (streaming), or near-resonant inertial waves[?].

A.10.6 A fluid-inspired kinematic time hypothesis (legacy variant)

Define $L \equiv \frac{1}{2} r_e$ with classical electron radius $r_e = e^2/(4\pi\epsilon_0 m_e c^2) \approx 2.82 \times 10^{-15} \text{ m}$. With $\omega = 2u_\theta/C_e$, define

$$\alpha_f \equiv \frac{\omega L}{c} = \frac{r_e}{c C_e} u_\theta, \quad \frac{d\tau}{dt} = \sqrt{1 - \alpha_f^2}. \quad (\text{A.28})$$

For $\alpha_f \ll 1$, $d\tau/dt \approx 1 - \frac{1}{2}\alpha_f^2$. Quadratic parity: angle reverses; time deficit does not.

Part III: Skymionic Photon Emission from Knotted Swirl Sources

A.10.7 From optical skyrmions to VAM/SST topology

Optical skyrmions built from LG modes with opposite circular polarizations carry integer topological charge via Stokes field[?, ?, ?]:

$$N_{\text{sk}}^{(\text{ph})} = \frac{1}{4\pi} \int d^2 k_\perp \hat{S} \cdot (\partial_{k_x} \hat{S} \times \partial_{k_y} \hat{S}) \in \mathbb{Z}. \quad (\text{A.29})$$

Let $\hat{\omega} = \omega/\|\omega\|$; define vortex charge on Σ by

$$H_{\text{vortex}}[\hat{\omega}|\Sigma] \equiv \frac{1}{4\pi} \int_\Sigma \hat{\omega} \cdot (\partial_x \hat{\omega} \times \partial_y \hat{\omega}) dx dy. \quad (\text{A.30})$$

Topological inheritance. $N_{\text{sk}}^{(\text{ph})} = H_{\text{vortex}}[\hat{\omega}|\Sigma]$.

A.10.8 Projection law: from swirl to Stokes

Let $\mathbf{p}(\mathbf{r}) = p_0(\mathbf{r}) \hat{\omega}_\perp(\mathbf{r})$ on Σ . Far-field Jones amplitude in mode space (LG basis $u_{p,\ell}$, helicity $\sigma = \pm 1$):

$$A_{p\ell\sigma} = \int_{\Sigma} (\hat{e}_\sigma^* \cdot \mathbf{P}_\perp \hat{\omega}_\perp) u_{p,\ell}(\mathbf{r}) e^{i\Phi(\mathbf{r})} d^2r, \quad (\text{A.31})$$

with $\mathbf{P}_\perp = \mathbf{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^\top$ and a phase Φ . Stokes field computed from $\mathbf{E}$ yields (A.29).

A.10.9 VAM/SST radiative vertex: OAM additivity and chirality

Per-photon OAM in SPDC conserves ℓ additively[?, ?]; adopt

$$\ell_{\text{src}} = \sum_{j=1}^n \ell_j \quad (n\text{-photon channel}). \quad (\text{A.32})$$

Swirl chirality sets photon helicity: ccw $\Rightarrow \sigma = +1$, cw $\Rightarrow \sigma = -1$.

A.10.10 Frequency and energy scale

$$\Omega_0 \equiv \frac{C_e}{r_c}, \quad E_0 = \hbar\Omega_0. \quad (\text{A.33})$$

Numerically (given values): $\Omega_0 \simeq 7.77 \times 10^{20} \text{ s}^{-1}$, $E_0 \simeq 0.511 \text{ MeV}$.

Quantized line set. Let

$$\omega_{m\ell} = m \Omega_0 + \delta\omega_\ell, \quad E_{m\ell} = \hbar\omega_{m\ell} = m E_0 + \delta E_\ell. \quad (\text{A.34})$$

A.10.11 Predictions

- **P1 (topological spectroscopy):** $N_{\text{sk}}^{(\text{ph})} = H_{\text{vortex}}$ within a fixed mode family.
- **P2 (chirality–helicity):** flip swirl chirality $\Rightarrow$ flip photon helicity; OAM additivity (A.32) intact.
- **P3 (robustness):** $N_{\text{sk}}^{(\text{ph})}$ invariant under smooth, linear propagation.
- **P4 (Purcell tuning):** LDOS changes reshape texture without changing integer charge.
- **P5 (line assignments):** $m = 1$ near $m_e c^2$; absence of a strong ladder at $n \times 0.511 \text{ MeV}$ constrains δE_ℓ and higher- m couplings.

A.11 Minimal experimental roadmap

A.11.1 E1: Re-analyze single-photon skyrmion data

Express measured Stokes fields via a reconstructed $\hat{\omega}_\perp$; test $N_{\text{sk}}^{(\text{ph})} = H_{\text{vortex}}$, and OAM/helicity control.

A.11.2 E2: Swirl-based emitters at non-optical frequencies

Macroscopic swirl sources coupled via electro-/magneto-/piezo-optic effects should reproduce topological mapping independent of absolute energy scale.

A.11.3 E3: High-energy cross-checks

Consistency with electron mass/QED precision and absence of extra stable lines near multiples of 0.511 MeV.

Global Conclusion

Part I. Inertial-wave packet with $c_{g,z}$ explains delayed axial "push-pull"; $t_{\text{arr}} \propto \Omega^{-1}$ and $\eta_{\text{max}} \propto \Omega$. Time-rate hypotheses are even in speed.

Part II. Vertical forcing produces opposite-sign vorticity above/below; linear reversibility gives zero net angle over a symmetric cycle.

Part III. Skyrmionic photon textures inherit vortex topology; OAM additivity and chirality-helicity mapping follow; $\Omega_0 = C_e/r_c$ sets a kinematic scale with numerical validation.

Analogy (10-year-old). Knots in a river can flick patterns into light; how they're tied (knot type) decides the pattern, not how loudly you splash.

Acknowledgments. Fluid/wave results: [?, ?, ?, ?, ?, ?]. Topological/OAM optics: [?, ?, ?, ?, ?]. Relativistic/causality constraints: [?, ?, ?, ?].